

Unit 39: Trigonometric Substitution

1. The idea, in plain words

Some integrals hide an expression like $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, or $\sqrt{x^2 - a^2}$ — and no amount of ordinary substitution or algebra will tame that square root. The fix is a clever trick: replace x with a trig function of a new variable θ , chosen specifically so the square root collapses using the Pythagorean identity.

Picture why this works, using the $\sqrt{a^2 - x^2}$ case as an example. If you let $x = a \sin \theta$, then:

$$a^2 - x^2 = a^2 - a^2 \sin^2 \theta = a^2(1 - \sin^2 \theta) = a^2 \cos^2 \theta$$

Suddenly, the square root of that is just $a \cos \theta$ — clean, no root left at all! The three standard substitutions each target one specific “shape” of expression, chosen so the Pythagorean identity ($\sin^2 \theta + \cos^2 \theta = 1$, or its rearranged cousins) knocks out the square root every time.

After substituting, you’re left with an integral purely in terms of θ — often one you can handle with the trig integral techniques from Unit 38, or with the classic formula $\int \sec \theta d\theta = \ln|\sec \theta + \tan \theta| + C$ (a standard result worth just remembering, similar to how you memorized the basic trig antiderivatives).

The last step is always the same: convert your answer back from θ to x . Since θ was defined by an equation like $x = a \sin \theta$, you can picture θ as an angle in a right triangle, and read off whatever other trig ratios you need (like $\cos \theta$ or $\tan \theta$) directly from that triangle — the exact same technique from Unit 35.

2. Toolbox

The three substitution patterns:

If you see...	Substitute...	It becomes...
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$a \cos \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$a \sec \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$a \tan \theta$

Don’t forget to substitute dx too:

$$x = a \sin \theta \Rightarrow dx = a \cos \theta d\theta \quad x = a \tan \theta \Rightarrow dx = a \sec^2 \theta d\theta \quad x = a \sec \theta \Rightarrow dx = a \sec \theta \tan \theta d\theta$$

A key formula you’ll need for the $\tan \theta$ and $\sec \theta$ cases:

$$\int \sec \theta d\theta = \ln|\sec \theta + \tan \theta| + C$$

Converting back from θ to x — the reference triangle: draw a right triangle matching the substitution equation (e.g., for $x = a \sin \theta$, since $\sin \theta = \frac{x}{a}$, label opposite = x , hypotenuse = a ; find the missing side with the Pythagorean theorem), then read off whatever ratio you need directly from the triangle.

The full procedure:

1. Identify which of the three patterns matches your integral.
2. Substitute for x and dx ; simplify the square root using the identity.
3. Integrate the resulting expression in θ .
4. Convert the answer back to x , using a reference triangle for any leftover trig ratios.

3. Common mistakes

- Choosing the wrong substitution pattern. Double-check whether you have $a^2 - x^2$ (sine), $a^2 + x^2$ (tangent), or $x^2 - a^2$ (secant) — these are genuinely different shapes and need different substitutions.
- Forgetting to substitute dx as well as the square root expression. Both pieces of the integral need to be rewritten in terms of θ .
- Forgetting to convert the final answer back to x . An answer left in terms of θ isn't finished — always draw the reference triangle and substitute back.
- Mislabeling the reference triangle. Match the sides to the specific substitution equation you used (e.g., for $x = a \sec \theta$, secant is hypotenuse-over-adjacent, so hypotenuse = x , adjacent = a) — mixing this up gives you the wrong ratios.
- Forgetting the formula for $\int \sec \theta d\theta$ and trying to reinvent it from scratch — it's a standard result worth memorizing directly.

4. Problem Set

Warm-up

1. Which trig substitution would you use for $\int \frac{dx}{\sqrt{9-x^2}}$?
2. Which trig substitution would you use for $\int \frac{dx}{\sqrt{x^2+16}}$?
3. Which trig substitution would you use for $\int \frac{dx}{\sqrt{x^2-25}}$?
4. $\int \frac{dx}{\sqrt{4-x^2}}$ using $x = 2 \sin \theta$
5. $\int \frac{dx}{\sqrt{9-x^2}}$ using $x = 3 \sin \theta$
6. $\int \sqrt{1-x^2} dx$ using $x = \sin \theta$

Standard

7. $\int \frac{x^2}{\sqrt{4-x^2}} dx$ using $x = 2 \sin \theta$
8. $\int \frac{dx}{\sqrt{x^2+4}}$ using $x = 2 \tan \theta$
9. $\int \frac{dx}{\sqrt{x^2+9}}$ using $x = 3 \tan \theta$
10. $\int \frac{dx}{(x^2+4)^{3/2}}$ using $x = 2 \tan \theta$
11. $\int \frac{dx}{\sqrt{x^2-4}}$ using $x = 2 \sec \theta$
12. $\int \frac{dx}{x^2 \sqrt{x^2-9}}$ using $x = 3 \sec \theta$

Challenge

13. $\int \sqrt{9-x^2} dx$ using $x = 3 \sin \theta$
14. $\int x^2 \sqrt{9-x^2} dx$ using $x = 3 \sin \theta$
15. $\int \frac{dx}{\sqrt{x^2+16}}$ using $x = 4 \tan \theta$
16. $\int \frac{dx}{(x^2+9)^{3/2}}$ using $x = 3 \tan \theta$
17. $\int \frac{dx}{x^2 \sqrt{x^2-4}}$ using $x = 2 \sec \theta$

Applied

18. Find $\int_{-2}^2 \sqrt{4-x^2} dx$, and check your answer against the known area formula for a semicircle.
19. Find $\int_0^3 x^2 \sqrt{9-x^2} dx$.
20. Find $\int_0^4 \frac{dx}{\sqrt{x^2+16}}$.
21. Find $\int_3^{3\sqrt{2}} \frac{dx}{x^2 \sqrt{x^2-9}}$.